

TASK 1: Hangman Game (with Execution Output)

CodeAlpha Python Programming Internship • Project Submission

1. Source Code Implementation

```
import random

def hangman_game():
    words = ["python", "coding", "alpha", "intern", "script"]
    secret_word = random.choice(words)
    guessed_letters = []
    incorrect_guesses = 0
    max_incorrect = 6

    print("--- Welcome to Hangman Game ---")

    while incorrect_guesses < max_incorrect:
        display_word = ""
        for letter in secret_word:
            if letter in guessed_letters:
                display_word += letter + " "
            else:
                display_word += "_ "

        print(f"\nWord: {display_word.strip()}")
        print(f"Incorrect guesses left: {max_incorrect - incorrect_guesses}")

        if "_" not in display_word:
            print(f"🎉 Congratulations! You guessed the word correctly: {secret_word}")
            break

        guess = input("Guess a letter: ").lower().strip()
        if len(guess) != 1 or not guess.isalpha():
            print("Please enter a single valid letter.")
            continue

        if guess in guessed_letters:
            print("You already guessed that letter. Try another one.")
            continue

        guessed_letters.append(guess)
        if guess in secret_word:
            print(f"Good job! '{guess}' is in the word.")
        else:
            print(f"Oops! '{guess}' is not in the word.")
            incorrect_guesses += 1

    else:
        print(f"\nGame Over! You ran out of guesses. The word was: {secret_word}")

if __name__ == "__main__":
    hangman_game()
```

2. Expected Terminal Output

```
--- Welcome to Hangman Game ---
```

```
Word: _ _ _ _ _
```

```
Incorrect guesses left: 6
```

```
Guess a letter: p
```

```
Good job! 'p' is in the word.
```

```
Word: p _ _ _ _
```

```
Incorrect guesses left: 6
```

```
Guess a letter: x
```

```
Oops! 'x' is not in the word.
```

```
Word: p _ _ _ _
```

```
Incorrect guesses left: 5
```

```
Guess a letter: y
```

```
Good job! 'y' is in the word.
```

```
Word: p y _ _ _
```

```
Incorrect guesses left: 5
```

```
Guess a letter: t
```

```
Good job! 't' is in the word.
```

```
Word: p y t _ _
```

```
... [Game Progresses] ...
```

```
Word: p y t h o n
```

```
Incorrect guesses left: 5
```

```
🎉 Congratulations! You guessed the word correctly: python
```